ProvGuard: Confusion-Selected Hard Negatives for Reducing Prompt-Injection Over-Defense under Distribution Shift

Anonymous Authors

***Abstract*—Prompt-injection detectors can reject benign requests that resemble attacks despite high in-distribution accuracy. Hard-negative augmentation may reduce this over-defense on curated stress examples yet transfer poorly to externally sourced prompts. We study whether the provenance and selection of benign augmentation data explain this gap. ProvGuard ranks naturally occurring Dolly and OpenAssistant prompts by the mean injection probability of three baseline detectors, applies lexical leakage controls and two-pass automated semantic adjudication, and retrains the same DistilBERT classifier on the selected benign examples. We complete a five-seed comparison with an unaugmented baseline, curated hard negatives, and an equal-sized source/length-matched Random-Real control, plus nested size ablations. At a fixed threshold, external hard-benign false-positive rate falls from 20.67% to 4.23%, a 79.5% relative reduction. IID F1 changes from 98.15% to 98.26%, while attack recall changes from 98.71% to 98.69%. ProvGuard improves on Random-Real's 7.31% external false-positive rate, with a positive paired-bootstrap interval for the reduction; the aggregate mixed-set comparison remains inconclusive. Curated augmentation performs best on the curated-heavy aggregate, and domain-OOD F1 decreases under ProvGuard. These results support provenance-sensitive, target-distribution gains rather than universal superiority or a new detector architecture.**

***Index Terms*—prompt injection, over-defense, hard-negative mining, false positives, provenance, data-centric robustness**

I. Introduction

Prompt injection attempts to redirect a language-model application through instructions that violate its intended task or trust boundary. Indirect injection places such instructions in external content consumed by the application [1]. A detector can screen that content before execution, but a useful screen must also accept benign discussion of attacks, system prompts, and security controls. Rejecting a request merely because it contains attack-associated wording imposes a usability cost.

High in-distribution (IID) performance can conceal this cost. Our unaugmented detector attains 98.15% IID F1 yet flags 20.67% of the available externally sourced hard-benign prompts. Prior work documents over-defense from trigger-word bias [2] and sensitivity to benign-data provenance [3]. Curated hard negatives can teach a detector to accept curated stress templates without adequately covering naturally occurring requests. This motivates a data-selection question rather than a new architecture: *Can naturally occurring benign prompts that a prompt-injection detector finds attack-like reduce over-defense on externally sourced benign inputs more effectively than generic benign augmentation, without materially weakening attack detection?*

ProvGuard selects relatively detector-confusing verified-benign prompts from Dolly and OpenAssistant. All arms retain the same classifier, original validation split, optimization settings, and threshold. An equal-sized, source/length-matched Random-Real control measures what targeted selection adds beyond realistic benign data. A curated arm reveals whether gains depend on evaluation provenance.

The evidence is mixed in an informative way. Random-Real already reduces external false positives substantially; targeted selection yields a further 3.08-percentage-point reduction. Curated-HN instead achieves the lowest aggregate hard-benign rate on a population dominated by curated prompts. ProvGuard's advantage over Random-Real is supported on the targeted external subset, while the aggregate interval crosses zero. Our contributions are:

1) **A data-selection method:** ensemble-confusion ranking of naturally occurring prompts, with persisted lexical exclusion and recorded semantic validity decisions.
2) **A controlled study:** 30 DistilBERT runs across 5 seeds, comparing baseline, curated, matched random, and three nested ProvGuard sizes.
3) **A target-distribution result:** external hard-benign FPR decreases from 20.67% to 4.23%, and from Random-Real's 7.31%, while IID attack recall remains effectively unchanged.
4) **A qualified analysis:** paired uncertainty estimates, hardness distributions, error transitions, size sensitivity, and contrasting structural- and domain-OOD outcomes.

II. Related Work

A. Injection Detection and Over-Defense

Greshake *et al.* demonstrate how indirect injection crosses application trust boundaries [1]. DataSentinel formulates detection against adaptive injected prompts as minimax learning [4]. Such attacker-facing work addresses a different objective from accepting legitimate attack-adjacent requests; our study does not evaluate adaptive adversaries or end-to-end application security.

PIGuard introduces NotInject and a training strategy to reduce over-defense caused by trigger-word bias [2]. CAPTURE jointly examines attacks and benign cases in context-aware settings [5]. PIDS-Bench measures hard-benign, obfuscated, domain-, and structural-shift behavior and reports provenance-sensitive effects of curated augmentation [3]. We address its

externally matched augmentation question with a matched real-data control. We do not compare directly with published full-coverage detector scores: the available public-text evaluation has different coverage, and no state-of-the-art claim follows.

B. Hard Negatives and Data-Centric Robustness

Hard-example mining is established, including online mining in object detection [6]. Shortcut learning provides a general explanation for high IID accuracy coupled with distribution-dependent failures [7]. ProvGuard's contribution is the controlled application of confusion-selected *naturally occurring benign instruction data* to a specific provenance-sensitive failure, rather than invention of mining or proof of shortcut removal.

Dolly contains human-written instruction records [8]; OpenAssistant provides crowdsourced conversations [9], [10]. Human origin does not establish a benign injection label. We therefore separate corpus extraction from semantic adjudication. DistilBERT [11] remains the fixed primary architecture.

III. Method

A. Scoring and Training Objective

For input x, the classifier estimates $f_\theta(x) = P(y = 1 \mid x)$, where $y = 1$ denotes injection and $y = 0$ benign. Every evaluation uses $\hat{y} = \mathbb{I}[f_\theta(x) \geq 0.5]$. Selector models are the unaugmented baselines with seeds $S = \{13, 42, 123\}$. Define

$$h(x) = \frac{1}{|S|} \sum_{s \in S} f_{\theta_s}(x), \qquad H_K = \mathrm{TopK}_{x \in B_{\text{verified}}}\, h(x). \tag{1}$$

Ranking is descending, with candidate identifier breaking ties. Each H_K example receives benign label 0, and the same classifier is fine-tuned on

$$D_{\text{ProvGuard}} = D_{\text{PIDS train}} \cup H_K. \tag{2}$$

The predefined primary setting is $K = 1000$; nested $K = 250$ and 500 sets are ablations. A top-ranked example need not exceed the classifier threshold.

B. Candidate Preparation and Lexical Exclusion

Figure 1 summarizes the pipeline. We use Dolly's instruction field without concatenating responses or context. OASST1 candidates are nondeleted, English, root-level prompter messages from its training split. We exclude empty texts, fragments shorter than 20 characters, and explicit corruption markers; normalized exact duplicates are removed internally. Retained text keeps its original formatting.

All accessible text from the frozen training, validation, and evaluation files forms the exclusion reference. Exact comparison lowercases and collapses whitespace. A character-boundary TF–IDF representation with 3–5-grams is fitted *once* on the canonical union of reference and potential augmentation texts, then persisted. Candidates with cosine similarity ≥ 0.90 to a reference are removed. Construction and independent checks transform text with this same representation. Reference evaluation text is used solely for overlap exclusion, not as augmentation or a labeled selection signal; this unsupervised lexical representation does see the available benchmark text.

The 17535 cleaned real candidates lose 117 exact and 18 near overlaps, leaving 17400. The valid curated augmentation has 343 rows after the same policy. These controls prevent exact and high-similarity lexical overlap with available benchmark text but cannot guarantee complete semantic independence. Original benchmark files remain frozen.

C. Automated Semantic Validity

Two separately instantiated Codex reviewers read full candidate texts with numerical scores withheld. They label operational intent as BENIGN, INJECTION, or AMBIGUOUS. Benign security explanations, quotations, fiction, and imperatives remain eligible unless they direct the current model to override higher-priority instructions, reveal hidden prompts, bypass controls, or exfiltrate secrets. This is semantic adjudication, not a keyword ban or content-safety assessment.

Both passes must mark BENIGN for acceptance; either INJECTION rejects, and all other disagreement or uncertainty excludes. The passes review the top-ranked batch and additional independent random-control candidates: 2927 paired records yield 2923 BENIGN, 3 INJECTION, and 1 AMBIGUOUS decisions. The primary set reaches rank 1004 to retain 1000 examples. Scores are hidden but ranked batch order is visible. Separate histories do not eliminate correlated model error, the exact serving model version was not exposed, and agreement is not human validation. Full texts, decisions, reasons, and procedure prompts are recorded for audit.

D. Controls and Matching

Baseline uses no added data. Curated-HN uses 343 valid upstream constructed negatives. Random-Real draws 1000 verified-benign real prompts independently of hardness: a fixed seed-42 permutation within corpus and full-candidate character-length quartiles supplies the first eligible records for each treatment quota. Both real arms contain 429 Dolly and 571 OASST1 prompts. Their 213-example overlap is allowed, avoiding forced disjointness or easy-negative selection. Source/length matching does not establish syntactic equivalence. Curated-HN is a provenance control, not a size-matched mining control.

IV. Experimental Setup

A. Frozen Evaluation and Public Coverage

We use PIDS-Bench at repository revision `7984e2d` [12], with 27093 original training rows, 3989 validation rows, and 3918 IID test rows. The released hard-benign file has 1472 records, but 664 LMSYS texts are withheld; every aggregate HB FPR here uses only 808 accessible texts. These comprise 600 curated and 208 externally sourced examples (Dolly and OASST1). The external subset is the target distribution, not a previously unseen corpus: augmentation and evaluation share corpus provenance after lexical exclusion.

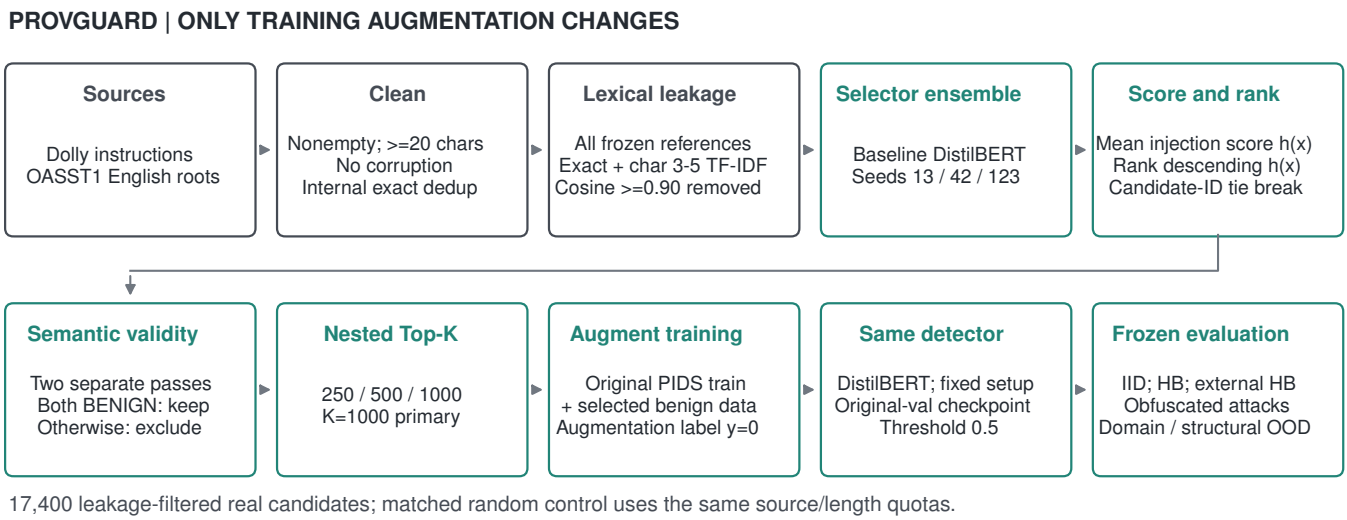


Fig. 1. ProvGuard changes training augmentation only. One persisted TF–IDF representation excludes lexical benchmark overlap; three baseline selectors rank candidate confusion; two separate automated adjudication passes establish operational benign eligibility. The fixed classifier and evaluation files are retained.

Other fixed evaluations contain 405 obfuscated attacks, 2000 domain-OOD examples, and 1998 structural-OOD examples. Subtype diagnostics remain in the artifacts. Stress-set FPR measures errors on constructed populations, not production prevalence. Published results using the withheld texts are not interchangeable with these denominators.

B. Training and Reproducibility

All six arms use seeds $\{13, 42, 123, 2024, 7777\}$, for 30 completed runs. We fine-tune the uncased base DistilBERT model with maximum length 256, 2 epochs, batch size 16, learning rate 5×10^{-5}, AdamW weight decay 0.01, and gradient-norm limit 1.0. The checkpoint with highest F1 on the *original* validation set is selected. Threshold 0.5 is fixed; no evaluation-set threshold or K tuning is performed. One RTX 4060 laptop GPU executes the study sequentially. Configurations, dataset hashes, model files, and per-example probabilities are retained for every run.

C. Metrics and Uncertainty

We report injection-class IID F1 and recall, benign-side FPR, obfuscated recall, domain-OOD F1, and structural-OOD FPR. Tables give five-seed means and sample SD. For comparator A, define $\Delta\text{FPR} = \text{FPR}_A - \text{FPR}_{\text{PG}}$, so positive deltas favor ProvGuard.

Aggregate intervals use 10000 crossed-bootstrap resamples: paired training seeds and common prompt identifiers are independently sampled with replacement, preserving method pairing and prompt dependence across seeds. A prompt's five predictions are not pooled as independent examples. Percentile intervals are pointwise 95% intervals, without aggregate multiplicity adjustment. Per-seed exact McNemar tests [13] and paired prompt bootstraps [14], with Holm correction across the per-seed tests, remain in the artifacts.

Before treatment training, the protocol required positive aggregate interval bounds on *both* HB endpoints for joint superiority over Random-Real. Preservation uses paired seed-bootstrap changes on the fixed IID set against predefined F1/recall loss bounds of $0.01/0.02$. These seed-based intervals do not quantify uncertainty over new attack populations. An external-only gain is therefore not reported as universal success.

V. Results

A. Baseline Over-Defense

Table I shows the gap between IID performance and benign usability. Despite 98.15% IID F1 and 98.71% attack recall, baseline HB FPR is 41.58%, and external HB FPR is 20.67%. Its substantial across-seed HB variation motivates paired uncertainty analysis rather than a single favorable run.

B. ProvGuard versus Baseline

ProvGuard reduces aggregate FPR to 13.84%: an absolute decrease of 27.75 percentage points (pp), or 66.7% relative. External FPR falls to 4.23%, a 16.44-pp decrease and 79.5% relative reduction (Fig. 2). Both aggregate intervals exclude zero (Table II).

IID F1 changes by +0.12 pp and attack recall by -0.03 pp, computed before rounding endpoint percentages. Paired seed-bootstrap intervals are $[-0.0006, 0.0033]$ for F1 and $[-0.0036, 0.0043]$ for recall, above the predefined lower preservation bounds. Obfuscated recall is 98.77% versus 98.57% at baseline. At this fixed operating point, the large benign-side gain is not accompanied by a material loss of measured IID attack detection.

C. Targeted Selection versus Random-Real

Random-Real itself reduces external FPR to 7.31%, demonstrating the value of realistic benign augmentation. ProvGuard

TABLE I

FINAL PRIMARY RESULTS: MEAN ± SAMPLE SD OVER FIVE SEEDS, FIXED THRESHOLD 0.5. HB FPR USES 808 PUBLIC TEXTS (600 CURATED + 208 EXTERNAL). PROVGUARD USES THE PREDEFINED $K = 1000$. BOLD DENOTES THE BEST OBSERVED MEAN AMONG THESE FOUR ARMS, NOT STATISTICAL SUPERIORITY.

Method	Aug. N	IID F1 ↑	Attack recall ↑	HB FPR ↓	External FPR ↓	Obf. recall ↑	Domain F1 ↑	Struct. FPR ↓
Baseline	0	0.9815 ± 0.0017	0.9871 ± 0.0047	0.4158 ± 0.1170	0.2067 ± 0.0350	0.9857 ± 0.0119	0.9033 ± 0.0409	0.7748 ± 0.1033
Curated-HN	343	0.9828 ± 0.0020	**0.9892** ± 0.0017	**0.0587** ± 0.0163	0.1846 ± 0.0341	**0.9921** ± 0.0011	**0.9131** ± 0.0348	0.5359 ± 0.1259
Random-Real	1000	**0.9840** ± 0.0009	0.9877 ± 0.0013	0.2015 ± 0.0638	0.0731 ± 0.0104	0.9891 ± 0.0028	0.8899 ± 0.0115	0.4356 ± 0.1309
ProvGuard	1000	0.9826 ± 0.0010	0.9869 ± 0.0016	0.1384 ± 0.0465	**0.0423** ± 0.0053	0.9877 ± 0.0039	0.8839 ± 0.0090	**0.3754** ± 0.1203

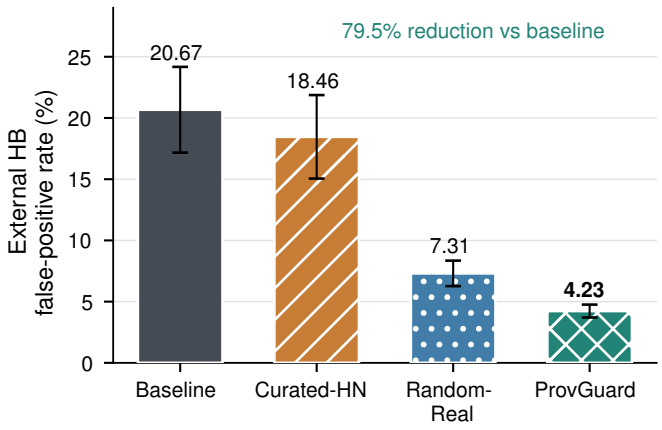


Fig. 2. External hard-benign FPR on 208 accessible prompts. Bars show five-seed means; error bars show sample SD, not confidence intervals. The equal-sized Random-Real control isolates the additional effect of confusion selection.

TABLE II

AGGREGATE CROSSED-BOOTSTRAP COMPARISONS WITH PROVGUARD. ΔFPR = COMPARATOR MINUS PROVGUARD; POSITIVE FAVORS PG. INTERVALS ARE POINTWISE 95%, OVER PAIRED SEEDS AND SHARED PROMPTS, WITH 10000 RESAMPLES. RATES ARE ON THE 0–1 SCALE.

Comparator	Subset	ΔFPR	95% CI	Interpretation
Baseline	Overall	+0.2775	[0.2007, 0.3512]	PG lower
Baseline	External	+0.1644	[0.1173, 0.2144]	PG lower
Random-Real	Overall	+0.0631	[−0.0050, 0.1408]	Inconclusive
Random-Real	External	+0.0308	[0.0096, 0.0548]	PG lower
Curated-HN	Overall	−0.0797	[−0.1267, −0.0379]	Curated lower
Curated-HN	External	+0.1423	[0.0933, 0.1933]	PG lower

lowers it further to 4.23%: 3.08 pp absolute and 42.1% relative. The crossed-bootstrap interval for this reduction, $[0.0096, 0.0548]$, is positive. All five paired seed differences favor ProvGuard on this endpoint.

On aggregate HB, the point improvement is 6.31 pp, but the interval $[-0.0050, 0.1408]$ includes zero. Thus the external comparison supports a targeted-distribution advantage, while the predefined joint novelty verdict remains *inconclusive*. Pointwise endpoint evidence must not be promoted to an adjusted, global superiority claim.

D. Provenance-Sensitive Transfer

Curated-HN has the lowest aggregate FPR, 5.87%, outperforming ProvGuard's 13.84%. This is a substantive result, not an exception to conceal. On the curated component alone, mean FPR is 0.0150 for Curated-HN versus 0.1717 for ProvGuard. Since 600 of 808 usable stress prompts are curated, the mixture rewards curated-style generalization.

The ordering reverses on external data: Curated-HN yields 18.46% FPR and ProvGuard 4.23%, a 14.23-pp reduction (77.1% relative). Both directional comparisons have intervals excluding zero. The contrast indicates provenance-sensitive augmentation effects. It does not establish that ranking alone explains the curated comparison, because the curated arm is smaller and differs in construction.

E. Augmentation Size

Figure 3(a) and Table III show non-monotonic behavior: $K = 250$ has the lowest aggregate HB FPR, 12.40%, whereas $K = 500$ has the lowest external FPR, 3.46%. At $K = 1000$, these are 13.84% and 4.23%. These are descriptive test outcomes; the primary configuration remains $K = 1000$. Increasing augmentation size changes composition and does not guarantee a better operating point.

F. OOD Behavior

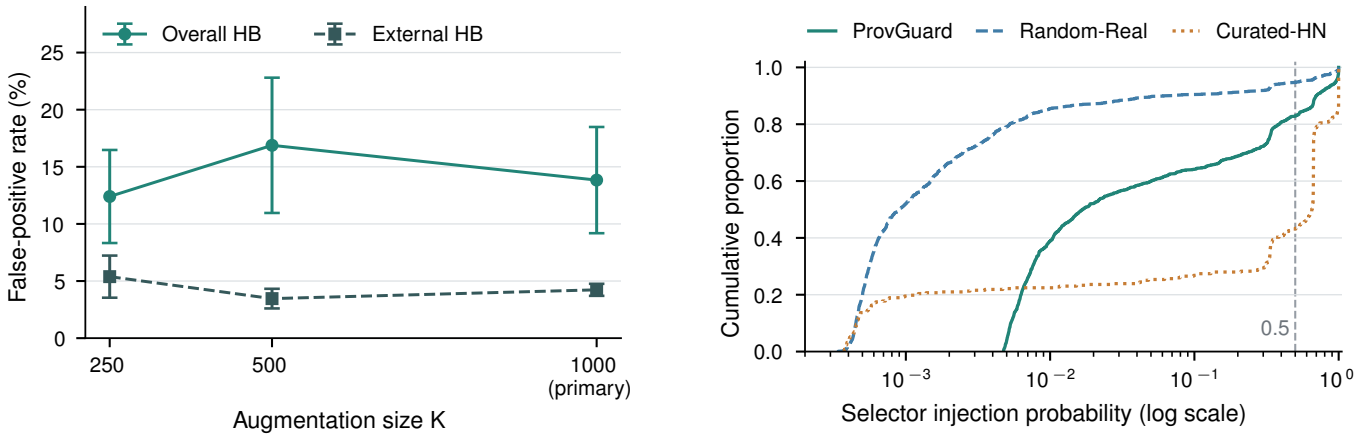
Structural-OOD benign FPR falls from 77.48% to 37.54%, while domain-OOD F1 falls from 0.9033 to 0.8839. The latter change is -1.94 pp. Structural benign-side improvement therefore coexists with a domain-shift cost, and does not establish universal OOD robustness. The structural set contains attacks as well as benign transforms; an FPR gain alone does not characterize its complete security behavior.

G. Hardness and Error Analysis

Selection changes the difficulty distribution (Fig. 3(b)). ProvGuard's mean selector score is 0.198910, versus 0.058424 for Random-Real; medians are 0.016738 and 0.000903. Only 172 of its 1000 examples exceed mean probability 0.5, versus 53 random examples. These are relatively confusing verified-benign examples, not uniformly misclassified inputs. This is a manipulation check, not independent evidence of generalization. Curated-HN is even harder by the same score (mean 0.493198) yet transfers worse externally, showing that score magnitude alone is insufficient.

TABLE III
NESTED AUGMENTATION-SIZE ABLATION: FIVE-SEED MEAN ± SAMPLE SD. $K = 1000$ REMAINS PRIMARY; NO SIZE IS SELECTED RETROSPECTIVELY FROM EVALUATION FPR. ALL RATES ARE ON THE 0–1 SCALE.

K	IID F1	Attack recall	HB FPR	External FPR	Obf. recall
250	0.9827 ± 0.0021	0.9854 ± 0.0047	0.1240 ± 0.0407	0.0538 ± 0.0184	0.9867 ± 0.0051
500	0.9824 ± 0.0005	0.9869 ± 0.0026	0.1688 ± 0.0592	0.0346 ± 0.0086	0.9891 ± 0.0028
1000	0.9826 ± 0.0010	0.9869 ± 0.0016	0.1384 ± 0.0465	0.0423 ± 0.0053	0.9877 ± 0.0039



(a) Augmentation-size sensitivity

(b) Selector-score distributions

Fig. 3. (a) Performance varies non-monotonically with augmentation size; $K = 1000$ was predefined as primary. Error bars are five-seed sample SD. (b) Empirical CDFs of baseline ensemble scores; the log-scale axis exposes the skewed benign distributions. Curated-HN has 343 rows; both real arms have 1000. The gray line marks $h = 0.5$.

Aligned aggregate HB predictions show an average of 245.0 baseline false positives corrected and 20.8 new false positives introduced per seed. New errors are retained in the artifacts. Released curated taxonomies and ordered lexical proxies categorize remaining errors; these exploratory automated categories are distinct from validity labels and cannot establish a learned semantic mechanism.

VI. DISCUSSION

The controlled comparison separates two effects. Real benign data already reduces mismatch: Random-Real achieves much of the external gain. Confusion selection then adds an endpoint-specific improvement under equal augmentation size and matched corpus/length quotas. This is consistent with addressing underrepresented benign decision-boundary regions, but we do not measure intent representations or establish a causal shortcut-removal mechanism.

The aggregate/external reversal against Curated-HN is central. The mixture is curated-heavy, and curated augmentation resembles that component's construction. ProvGuard better covers external requests but leaves more curated false alarms. Reporting only the mixture would prefer Curated-HN and hide the targeted-domain benefit; reporting only external performance would hide the opposite cost. Both views are necessary.

Aggregate seed variation and heterogeneous provenance responses help explain why the matched random comparison remains uncertain even when its external interval is positive. Source/length matching controls selected marginals, not all language structure. The non-monotonic size ablation likewise favors treating composition as consequential rather than assuming more negatives always help.

Exploratory source-only augmentation checks ($K = 250$; 3 seeds per arm) further show asymmetric behavior. Dolly augmentation yields FPRs of 0.0000 on 68 Dolly prompts and 0.2095 on 140 OASST prompts; OASST augmentation yields 0.0196 and 0.0619, respectively. Only augmentation provenance is isolated: the original PIDS training split is retained, so this is not fully held-out-corpus training. These small, descriptive checks support inspecting provenance slices, not a general cross-corpus guarantee.

These results support a practical lesson: validate benign provenance slices alongside attack recall before choosing augmentation. They do not support replacing trust-boundary controls with a classifier or treating stress FPR as a production rate. Generalization to new corpora, production traffic, and stronger architectures remains unestablished.

VII. LIMITATIONS

The external subset contains only 208 prompts; 664 LMSYS texts cannot be evaluated. Dolly and OASST1 are shared candidate/evaluation corpora, not a representative sample of production traffic. Lexical exclusion cannot ensure semantic independence. Automated adjudicators can share errors, are not human annotators, and lack an exposed exact model identifier. A 172-row high-score queue is provided for a future human audit; no completed human labels are claimed.

The primary architecture is DistilBERT: an optional DeBERTa attempt failed technically and supplies no transfer evidence. Five seeds limit uncertainty resolution; the selector ensemble and one random augmentation draw are fixed across those seeds. Additional random-sample checks lack completed reviews. Curated-HN uses 343 examples and is not size matched. The joint Random-Real comparison is inconclusive, domain-OOD F1 decreases, and source/length matching does not guarantee syntactic matching. No adaptive-attacker, unseen-transform, or end-to-end security claim is made.

VIII. Conclusion

ProvGuard shows that the provenance and selection of benign augmentation materially shape prompt-injection over-defense. Confusion-selected real hard negatives lower external HB FPR from 20.67% to 4.23% while retaining measured IID attack recall. They improve over matched Random-Real on the targeted external subset, but the aggregate comparison remains inconclusive and curated augmentation wins the curated-heavy mixture. Non-monotonic size effects and a domain-OOD cost further qualify the result. The contribution is a reproducible data-selection study with distribution-specific benefits, rather than a universally stronger detector.

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